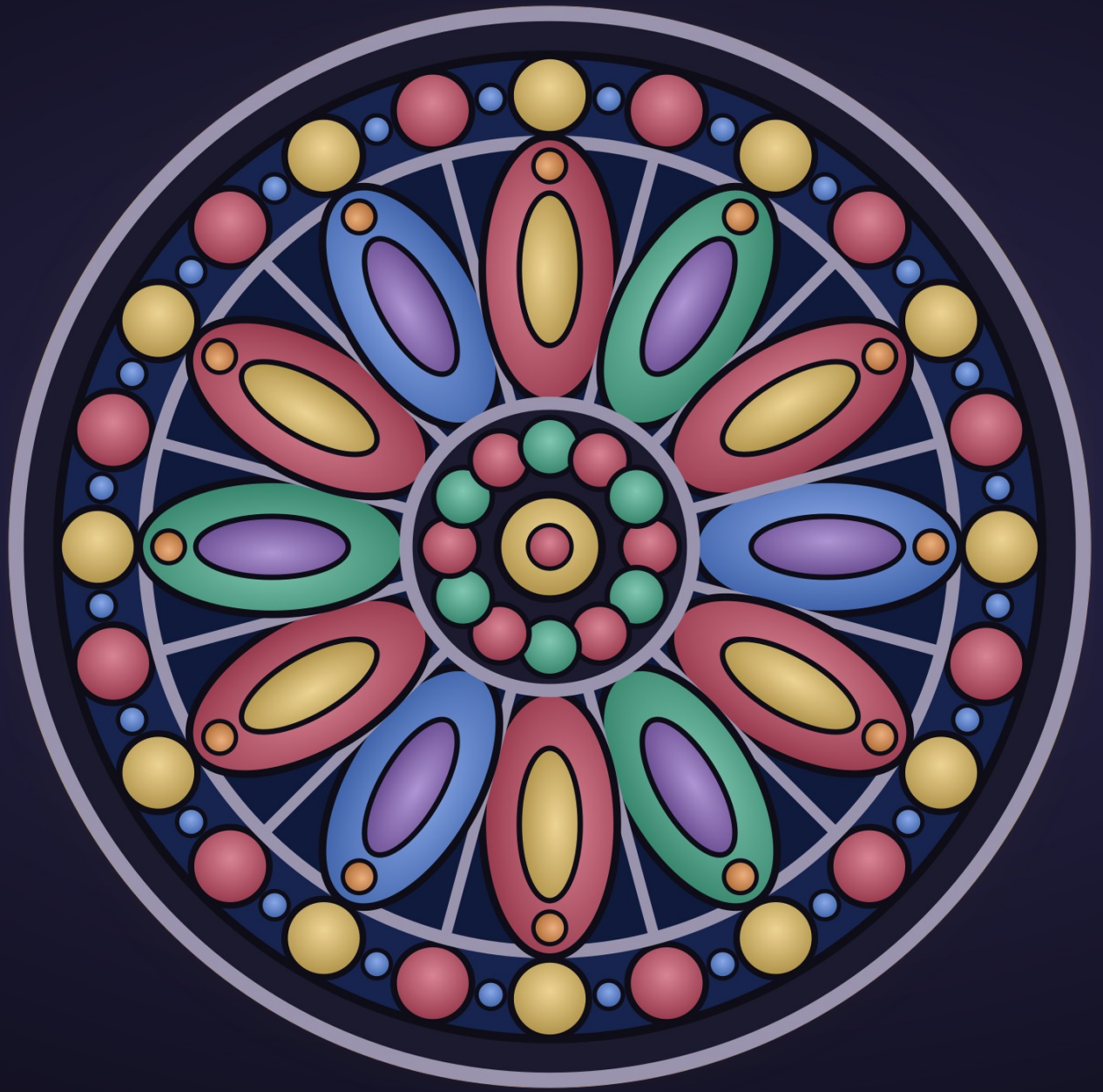
FIRST-YEAR ARCHITECTURE · HISTORY

Gothic Architecture

Stone, light and the race towards heaven



Presented by [Your name] · [Date]

Why does it still give us chills?

1140

Abbot Suger starts rebuilding Saint-Denis, near Paris

≈ 400

years of Gothic building across Europe

48 m

Beauvais: the highest Gothic vault ever built

632

years to finish Cologne Cathedral (1248–1880)

TODAY'S JOURNEY

1 Origins & faith

2 Structure

3 Space & making

4 Legacy

Four centuries in four chapters

1140–1200

Early Gothic

Born near Paris

- Saint-Denis (1140)
- Sens
- Notre-Dame de Paris (1163)

1200–1240

High Gothic

Taller, lighter, bolder

- Chartres
- Reims
- Amiens

1240–1350

Rayonnant

Walls dissolve into glass

- Sainte-Chapelle (1248)
- Westminster Abbey
- Cologne (1248)

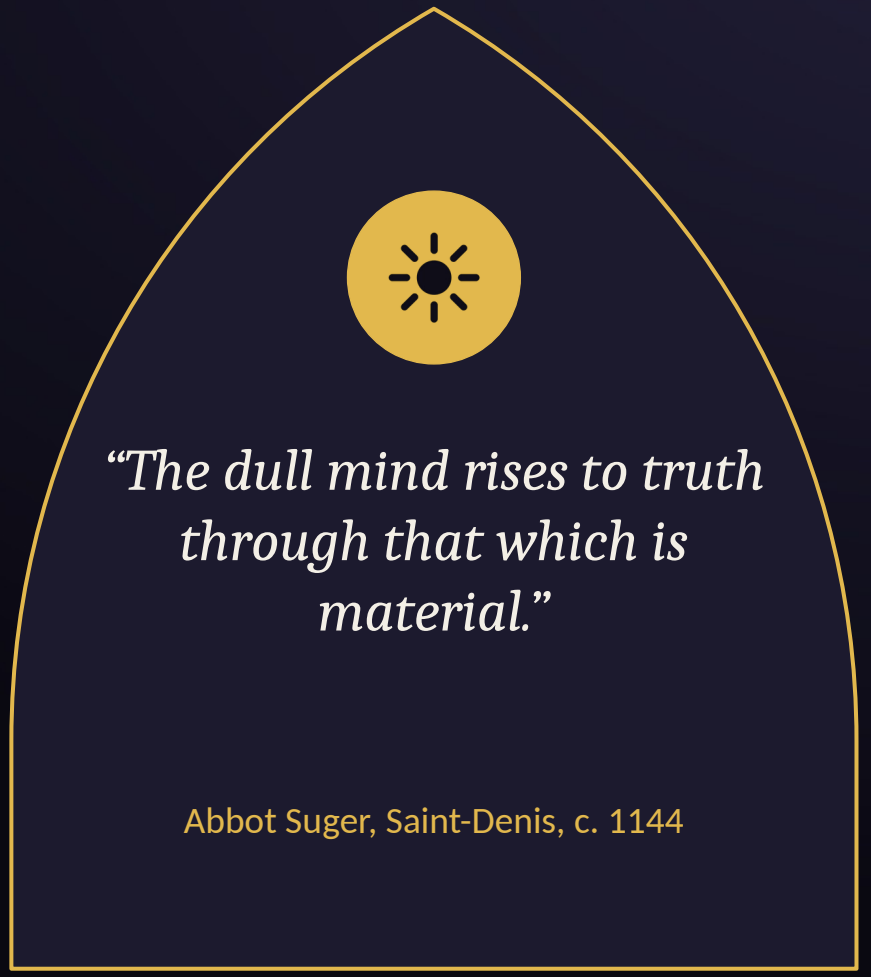
1350–1520

Late Gothic

Flames and fans

- Flamboyant tracery
- Fan vaults, England
- After the Black Death (1347)

Built for God: a heaven made of stone



Light is divine

God is pure light. Windows, gold and jewels lift the soul towards heaven.

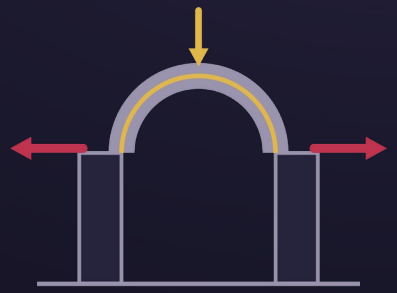
Mary and the pilgrims

Notre-Dame means “Our Lady”. Relics drew crowds, and their gifts paid the builders.

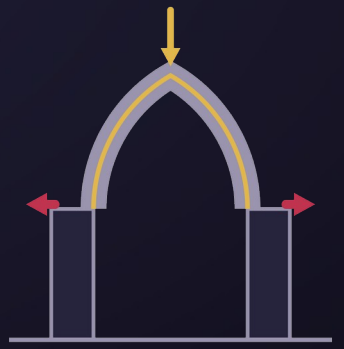
Power and pride

Bishops, kings and cities competed. A taller cathedral meant a more important town.

From fortress-church to lantern



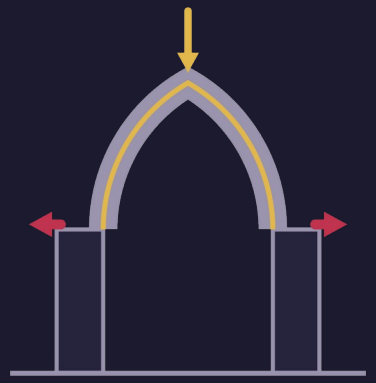
ROMANESQUE
c. 1000–1150



GOTHIC
c. 1140–1500

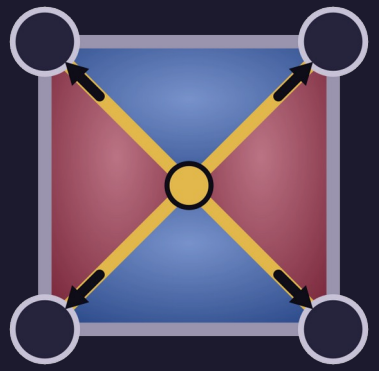
	ROMANESQUE	GOTHIC
ARCH	Round	**Pointed**
VAULT	Heavy barrel or groin vault	**Light ribbed vault**
WALLS	Thick, load-bearing mass	**Thin skin on a stone skeleton**
WINDOWS	Small and few	**Huge, filled with coloured glass**
FEELING	Solid, protective, earthly	**Soaring, bright, heavenly**

Three ideas that changed everything



1 Pointed arch

Steeper sides push the load down, not out. Any width can reach any height.



2 Ribbed vault

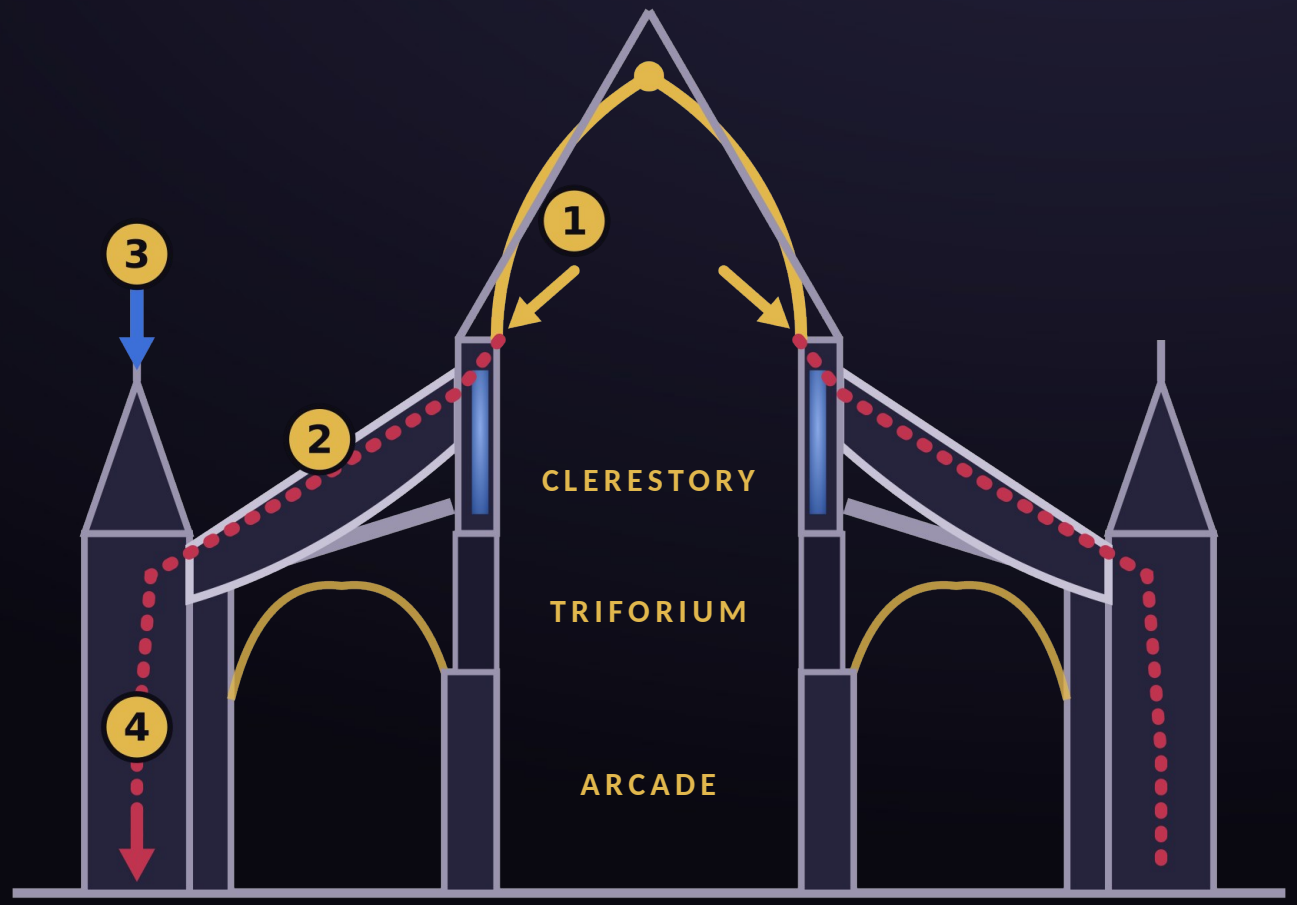
Stone ribs form a skeleton with thin panels between. Loads flow to the piers.



3 Flying buttress

An arch outside the wall catches the roof's push and carries it to a heavy pier.

Follow the force



1 — The vault pushes

Heavy stone presses down and outwards at the top of the wall.

2 — The flying buttress catches it

A half-arch sends the thrust over the aisle.

3 — The pinnacle adds weight

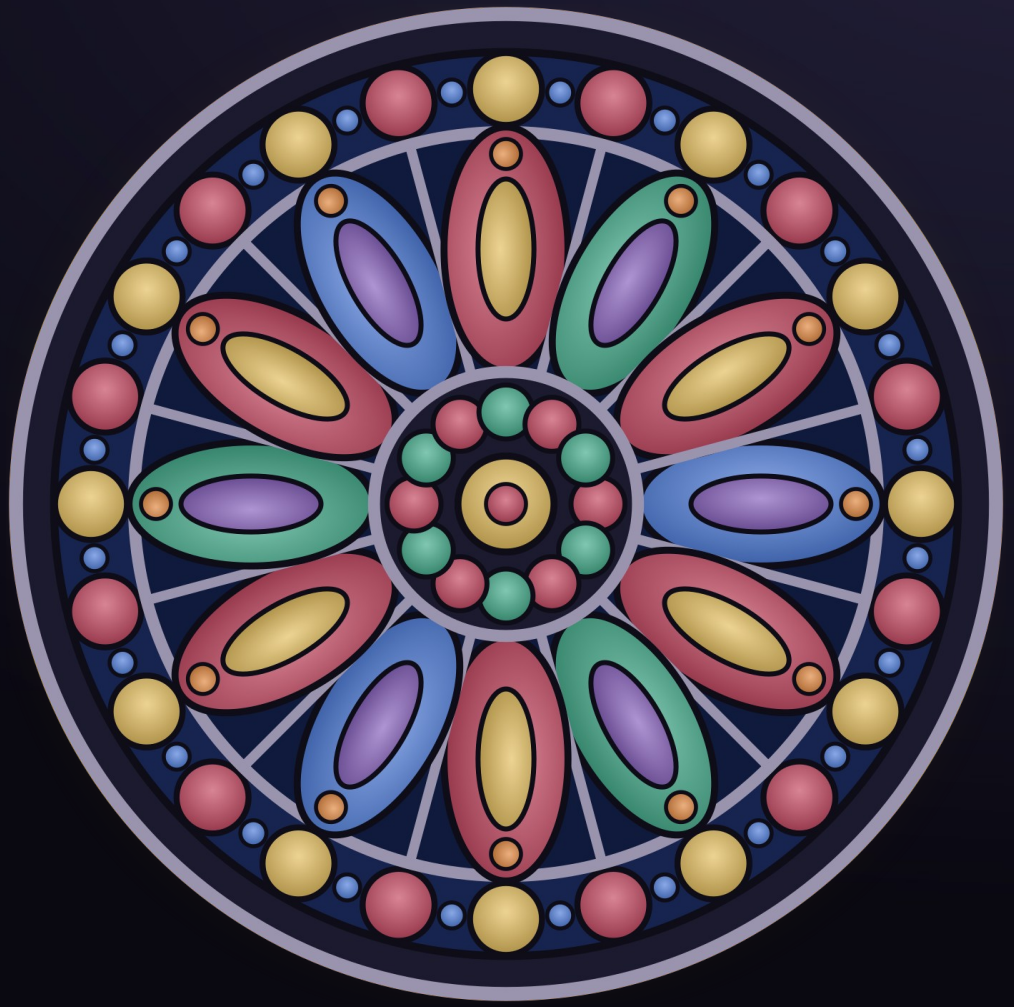
It keeps the line of force inside the pier.

4 — The pier reaches the ground

The wall itself carries almost nothing.

A cathedral is an exoskeleton.

Walls made of light



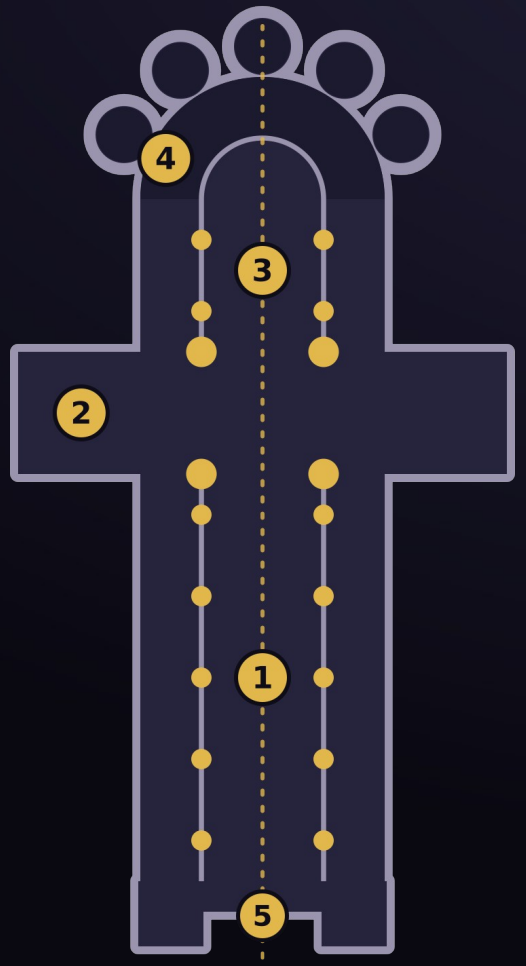
1,113 — Bible scenes in the Sainte-Chapelle windows, Paris

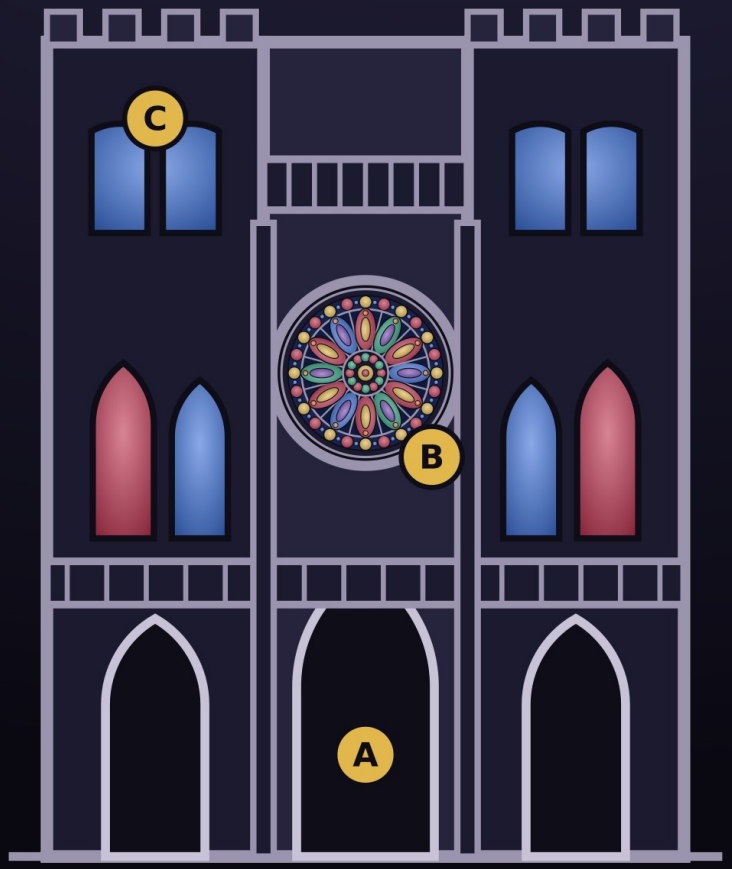
15 m — height of each window: glass replaces almost the whole wall

≈ 7 yrs — to build it: 1241–1248, for King Louis IX

Metal oxides colour the glass: cobalt blue, copper red.

Reading a cathedral





PLAN: A LATIN CROSS

1. **Nave & aisles** hall for the people
2. **Transept** arms of the cross
3. **Choir** clergy and altar, east
4. **Ambulatory** walkway to relic chapels
5. **West front** main entrance

FACADE

A. **Portals** carved stone stories
B. **Rose window** circle of glass
C. **Twin towers** bell towers

Built by generations, not by one genius



Fund

Bishops, kings, guilds and pilgrims pay. Money often ran out.



Design

A master mason draws with compass and geometry.



Carve

Each stone is cut on the ground and marked with the mason's sign.



Lift & set

Treadwheel cranes, scaffolds and wooden centering.



Wait

Decades, even centuries. Many builders never saw the end.

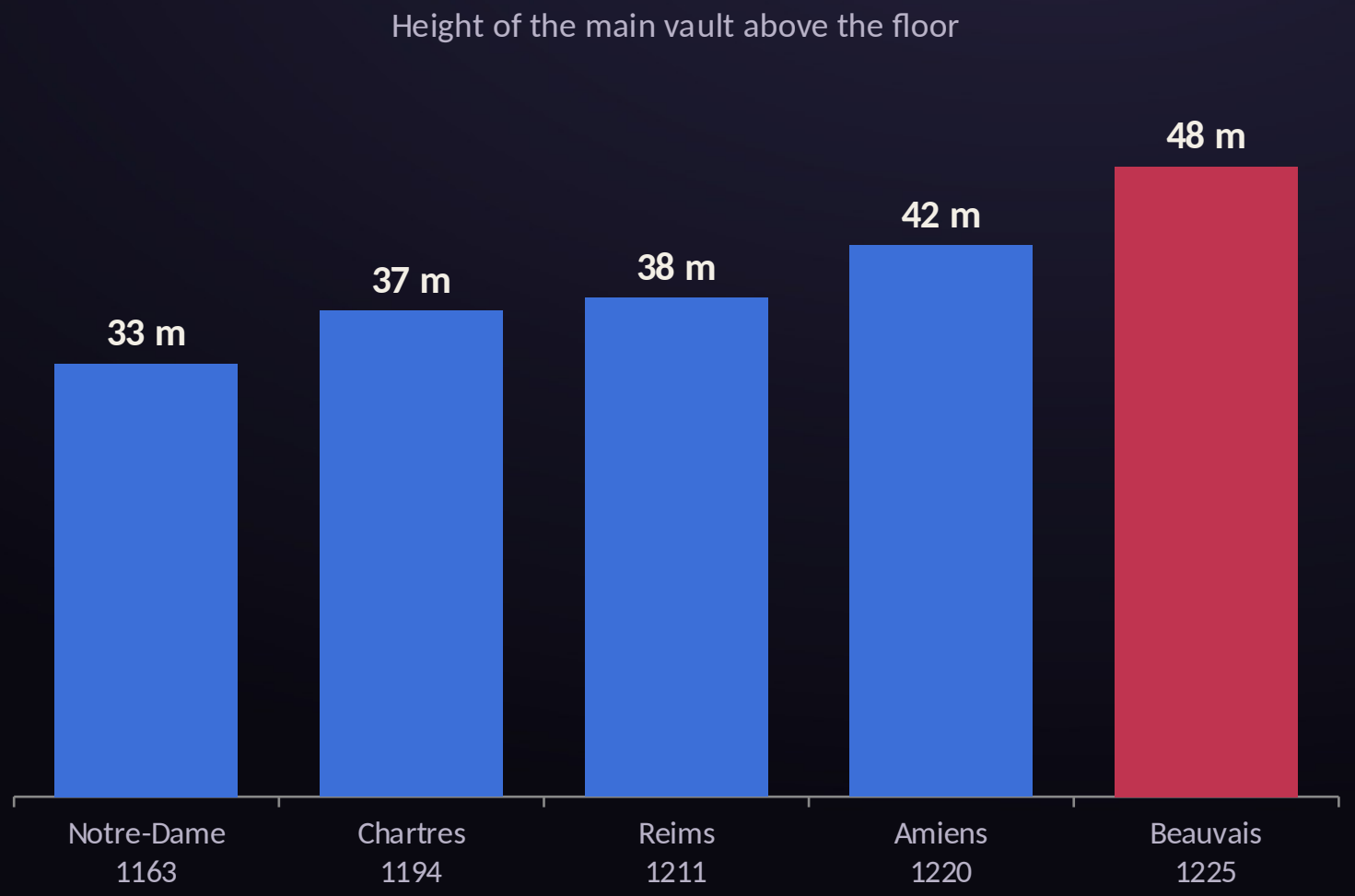
≈ 180 years

Notre-Dame de Paris, 1163 to about 1345

632 years

Cologne Cathedral, 1248 to 1880

The race to the sky



+45%

in about 60 years: each bishop wanted to beat the last.

1284

Beauvais' vault collapses, only 12 years after the choir was finished.

Trial & error

No structural theory yet: every failure taught the masons something.

Spread, decline and rebirth

1 Spread

Every country adds an accent

- England: Salisbury, Westminster Abbey
- Germany: Cologne
- Italy: Milan Duomo, in marble
- Spain: Seville, the largest Gothic cathedral

2 Decline

The Renaissance looks back to Rome

- Columns, domes and symmetry return
- Vasari calls the old style “Gothic”: barbarian, like the Goths
- The name began as an insult

3 Rebirth

Romantic love for the Middle Ages

- Houses of Parliament, London (1840)
- Viollet-le-Duc restores Notre-Dame
- Gaudí's Sagrada Família (1882)
- Notre-Dame: fire 2019, reopened 2024

What Gothic teaches an architect

1. **Structure can be beauty**

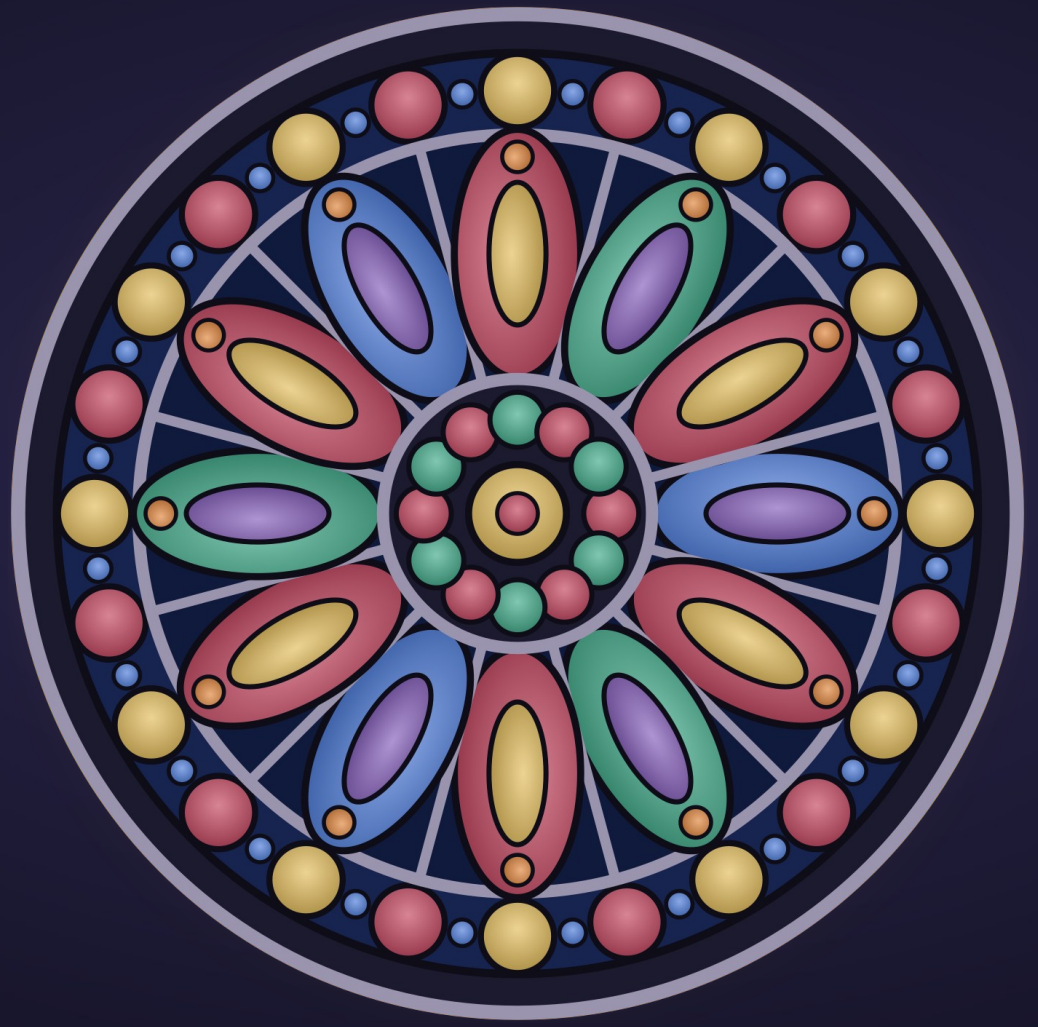
   The skeleton is the design. A frame with a glass skin is the ancestor of the skyscraper.

2. **Light is a material**

   Shape it, colour it, and the room changes with the hour.

3. **Big ideas take generations**

   Hundreds of hands and one shared dream.



Thank you. Questions?

Sources and further reading

Otto von Simson	The Gothic Cathedral: Origins of Gothic Architecture and the Medieval Concept of Order (1956)
Erwin Panofsky	Abbot Suger on the Abbey Church of St.-Denis and Its Art Treasures (1946)
Jean Gimpel	The Cathedral Builders (1958)
Georges Duby	The Age of the Cathedrals: Art and Society, 980–1420 (1976)
Online	Encyclopaedia Britannica and Wikipedia entries on Gothic architecture, Sainte-Chapelle, Beauvais Cathedral, Notre-Dame de Paris

All diagrams were drawn for this presentation.